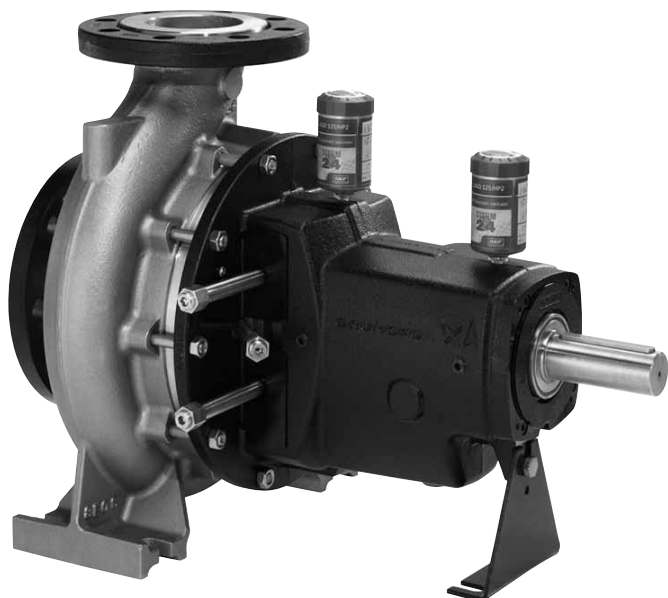


NKG

Double seal (tandem)

安装和使用说明书



中文 (CN)
安装和使用说明书 4

English (GB)
Installation and operating instructions 8

CONTENTS

	Page
1. Symbols used in this document	8
2. General information	8
3. Product description	8
4. Applications	8
5. Operating conditions	9
5.1 Flushing fluid	9
6. Installation	9
6.1 Flushing fluid connection	9
6.2 Flushing system	10
7. Start-up	11
8. Maintenance	11
9. Operation	11
9.1 Periods of inactivity and frost protection	11
10. Service	11
10.1 Service kits and service instructions	11
11. Disposal	11

**Warning**

Before beginning installation procedures, these installation and operating instructions should be studied carefully. The installation and operation should also be in accordance with local regulations and accepted codes of good practice.

1. Symbols used in this document

**Warning**

If these safety instructions are not observed, it may result in personal injury!

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment!

Note

Notes or instructions that make the job easier and ensure safe operation.

2. General information

These installation and operating instructions are a supplement to installation and operating instructions for NK, NKG.

3. Product description

This type of double seal consists of two shaft seals mounted in a tandem arrangement in a separate seal chamber or of a cartridge seal.

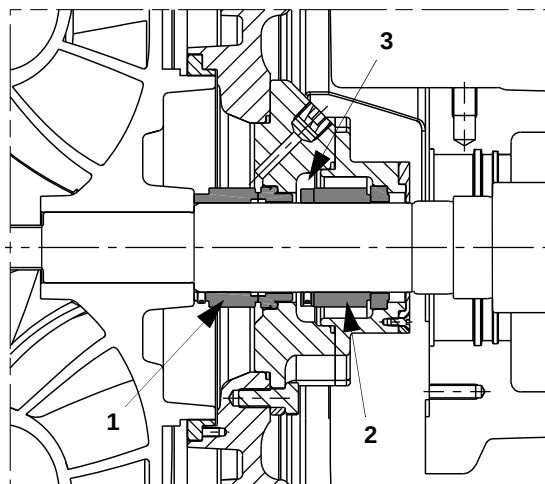


Fig. 1 Tandem seal arrangement consisting of two shaft seals

Pos.	Description
1	Primary shaft seal
2	Secondary shaft seal
3	Seal chamber, containing flushing fluid

Pumps with a tandem shaft seal arrangement require a flushing system providing the correct flushing fluid to the seal chamber.

The pressure in the seal chamber/cartridge seal **must** be lower than the pumped liquid pressure. Consequently, a small quantity of the pumped liquid will seep through the primary shaft seal and be mixed with the flushing fluid. The pumped liquid leaking via the primary seal will be flushed away by the flushing fluid.

As the primary shaft seal is in contact with liquid on both sides, there is no evaporation zone in the seal gap. This prevents build-up of crystallising deposits on the seal faces of the primary shaft seal. Deposits might otherwise lead to a failing primary shaft seal.

In hot applications, the flushing fluid additionally removes heat from the system both during operation and standstill, and thus cools the seal faces of the shaft seal.

4. Applications

The tandem shaft seal is the optimum solution in these cases:

- The pump is pumping crystallising liquids, e.g. a caustic soda solution. Build-up of crystals on the atmospheric side will be avoided.
- Lubrication and cooling of shaft seals is necessary during standstill.
- Monitoring of the leakage rate from the primary shaft seal is required.
- Prevention of air ingress from the atmospheric side is necessary (for liquids which react with atmospheric oxygen).
- The pump is operating with a negative inlet pressure (vacuum) of 0 - 0.69 bar compared to the pressure in the flushing chamber.
In this case the flushing fluid will provide the lubricating film for the primary shaft seal.

The tandem seal arrangement can handle a pumped liquid temperature up to 140 °C.

5. Operating conditions

The operating conditions stated in the installation and operating instructions for NK, NKG pumps also apply for tandem shaft seal pumps.

5.1 Flushing fluid

The flushing fluid must be clean.

Caution

The flushing fluid should be selected according to the application and must not chemically attack the materials of the pump and the shaft seal.

The flushing fluid must have a high boiling point, good lubricating and heat transmission properties.

The flushing-fluid flow should also be matched to the application (recommended flow rate 25-200 l/h).



Warning

Never connect the flushing-fluid supply directly to the public water supply system. Local regulations must be observed.

It is possible to connect the seal chamber/cartridge seal to a separate flushing-fluid tank. In such cases, the manufacturer's installation and operating instructions must be observed.

5.1.1 Maximum flushing-fluid pressure and temperature

The pressure of the flushing fluid must always be lower than the pressure of the pumped liquid.

The maximum permissible temperature of the flushing fluid depends on the rubber material of the secondary shaft seal.

Shaft seal	Max. permissible temperature of the flushing fluid [°C]
EPDM	+120
FFKM	+140
FKM	+90
FXM	+140
HNBR	+90

These values apply to water as flushing fluid. If other flushing fluids are used, contact Grundfos for the max. permissible temperature of the flushing fluid.

5.1.2 Flushing fluid evaporation point

The tandem shaft seal arrangement can handle a pumped liquid temperature up to 140 °C. In order to ensure proper liquid lubrication in the the seal gap, it is very important that the evaporation point of the flushing fluid is at least 10-15 °C higher than the temperature of the pumped liquid.

6. Installation

6.1 Flushing fluid connection

6.1.1 Tandem seal arrangement with two separate shaft seals in a seal chamber

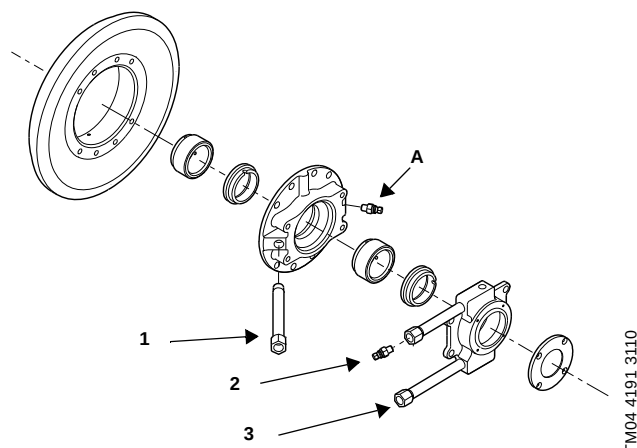


Fig. 2 Tandem seal arrangement with two separate shaft seals in a seal chamber

Pos.	Description
A	Vent screw for pumped liquid
1	Primary shaft seal pipe connection -3/8"
2	Secondary shaft seal pipe connection with vent screw for venting of seal chamber -3/8"
3	Secondary shaft seal pipe connection -3/8"

In this tandem shaft seal arrangement, the seal chamber has three connections. See fig. 2 and fig. 3. Pos. 1 leads to the pumped liquid side of the shaft seal, and pos. 2 and 3 lead to the seal chamber. All connections are close to the seal faces of the shaft seals.

Connections

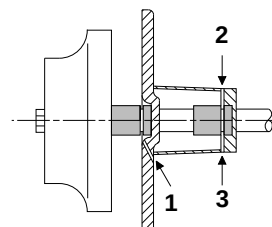


Fig. 3 Flow direction of flushing liquid

Dead-end solutions (see 6.2 Flushing system)

Use only connection 3 as inlet and 2 for venting. Connection 1 must be plugged.

Circulating solutions (see 6.2 Flushing system)

Use connection 3 as inlet and 2 as outlet. The flushing fluid lubricates and cools the secondary shaft seal. Connection 1 must be plugged.

Separate circulation/cooling of primary shaft seal

Use connection 1. A pipe can be fitted between the pump discharge and connection 1. Alternatively, an external supply can be fitted to connection 1.

Circulation or cooling of the primary seal is commonly used in the following situations:

- to avoid accumulation of particles in the seal face area
- to increase the pressure in the seal face area resulting in a higher evaporation point
- to vent the shaft seal area in order to avoid dry running
- to provide cooling of the shaft seal. (The friction between the seal faces results in a temperature increase of 10-20 °C above the pumped liquid temperature. In this case the pumped liquid itself can provide cooling).

6.1.2 Tandem seal arrangement with cartridge seal

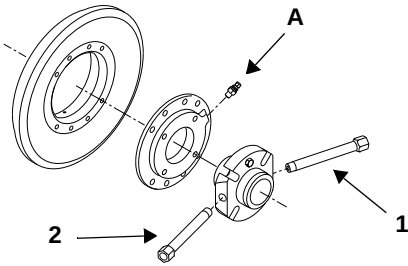


Fig. 4 Tandem seal arrangement with cartridge seal

Pos.	Description
A	Vent screw for pumped liquid
1	Cartridge seal inlet pipe connection -3/8"
2	Cartridge seal outlet pipe connections -3/8"

6.2 Flushing system

Typically, one of these methods of connecting the flushing fluid to the pump is used:

- circulation from a reservoir, see section 6.2.1
- dead end connection from a reservoir, see section 6.2.2
- external flushing fluid system, see section 6.2.3.

Common for these flushing solutions is that the pressure in the seal chamber is lower than the pressure of the pumped liquid around the shaft seal.

The flushing fluid acts as lubrication and cooling for the secondary seal and keeps a liquid phase for the primary seal in the seal chamber.

6.2.1 Circulation from a reservoir

Standard shaft seals: The seal chamber is connected to a reservoir by means of two pipes.

Both the primary and secondary shaft seal generates heat during operation. This heat energy is transferred to the flushing fluid. Due to natural circulation, the heated flushing fluid rises from the seal chamber to the reservoir where it cools down. The cooled-down flushing fluid returns to the seal chamber, lubricates and cools the seal faces.

After a period of time the flushing fluid in the reservoir must be replaced due to contamination from the pumped liquid.

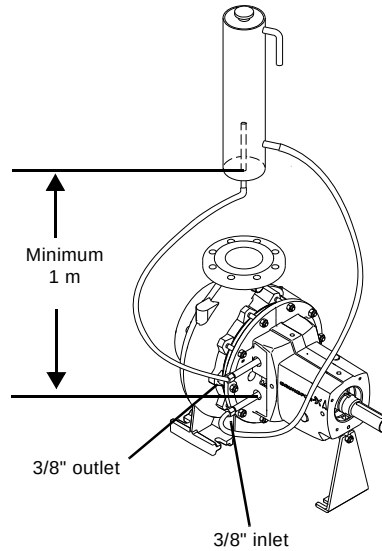


Fig. 5 Tandem seal arrangement with circulating flushing fluid

Circulation to a reservoir can also be forced by a separate pump.

Cartridge seal: A cartridge seal is connected to a reservoir by means of two pipes.

An internal pumping function in the shaft seal provides sufficient circulation of the flushing fluid to cool and lubricate the seal. The cartridge seal is self venting.

After a period of time the flushing fluid in the reservoir must be replaced due to contamination from the pumped liquid.

6.2.2 Dead-end connection from a reservoir

Standard shaft seals: The seal chamber is connected to a reservoir by means of a single pipe.

The flushing fluid lubricates the seal faces, but cools them less than in the case of circulation. After a period of time the flushing fluid in the reservoir must be replaced due to contamination from the pumped liquid.

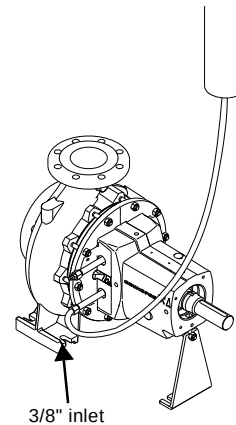


Fig. 6 Tandem seal arrangement with a dead-end flushing-liquid supply

The flushing fluid enters the seal chamber via a pipe from an elevated reservoir.

No heat is dissipated from the system.

Cartridge seal: A cartridge seal is **not** for use with a dead-end connection.

6.2.3 External flushing fluid systems

Standard shaft seals: The flushing fluid flushes the seal chamber and is led to a drain. The flushing fluid cools and lubricates the seal faces effectively.

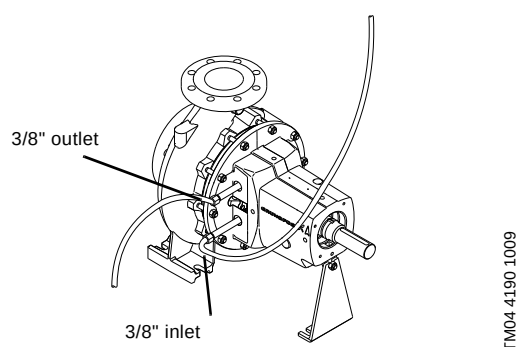


Fig. 7 Tandem seal arrangement with flushing fluid being led away to a drain

The pumped liquid leaking through the primary seal is washed away to the drain by the flushing fluid.

For external flushing fluid systems note these conditions:

- The flow rate of the flushing fluid should match the application (recommended flow rate 25-200 l/h. A temperature increase of 20 °C of the flushing fluid could also be used as a control parameter).
- The pressure of the flushing fluid must always be lower than the pressure of the pumped liquid.

Cartridge seal: The flushing fluid flushes the cartridge seal and is led to a drain. If the external flushing fluid inlet pressure drops, the internal pumping function of the cartridge seal provides sufficient circulation of the flushing fluid to cool and lubricate the seal faces. The cartridge seal is self venting.

Note

The flushing fluid supply must never be connected directly to the public water supply system. Local regulations must be observed.

7. Start-up

See also installation and operating instructions for the standard pump.

Caution

Before start-up, the seal chamber/cartridge seal and the pump must be completely filled with liquid and vented.

1. Connect the inlet and outlet of the seal chamber/cartridge seal by means of a suitable pipe or hose connection. See figures 2 and 4.
2. Start the pump, and check the direction of rotation. See the correct direction of rotation on the motor fan cover.

Caution

The seal chamber/cartridge seal must always be filled with flushing fluid during operation. Dry-running will destroy the shaft seal.

8. Maintenance

See installation and operating instructions for NK, NKG pumps.

9. Operation

To ensure optimum function and performance, make sure that the flow of flushing fluid is always sufficient, and check that the elevated tank never runs dry of flushing fluid.

Depending on the pumped liquid, the quality of the flushing fluid must be checked at regular intervals to ensure optimum operation.

9.1 Periods of inactivity and frost protection

Warning



Pay attention to the directions of the vent holes, and take care to ensure that the escaping liquid does not cause injury to persons or damage to the motor or other components.

When handling hot or hazardous liquids, special attention should be paid to the risk of injury caused by scalding hot liquid or hazardous liquid.

Local safety regulations must be observed.

Pump

For general guidelines, see installation and operating instructions for NK, NKG pumps.

Seal chamber

Unscrew the two connections (pos. 2 and 3, fig. 2).

The lower flushing-liquid connection functions as drain.

Cartridge seal

1. Unscrew the connections for flushing fluid (pos. 1 and 2, fig. 4).
2. Empty the cartridge seal. This can be done in two ways:
 - Blow out remaining liquid by means of an air gun or a similar device. The pressure must not exceed the pressure rating of the cartridge seal.
 - Flush the cartridge seal by means of a glycol-water solution, if applicable for the application. The pressure must not exceed the pressure rating of the cartridge seal.

10. Service

When ordering spare parts, always state the pump nameplate data to ensure that the correct parts are delivered.

If Grundfos is requested to service the pump, Grundfos must be contacted with details about the pumped liquid, etc. before the pump is returned for service. Otherwise Grundfos can refuse to accept the pump for service.

10.1 Service kits and service instructions

See www.grundfos.com (WebCAPS), WinCAPS or Service Kit Catalogue.

11. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana km. 37.500 Lote
34A
1619 - Garin
Pcia. de Buenos Aires
Phone: +54-3327 414 444
Telefax: +54-3327 411 111

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb
Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomsesteenweg 81-83
B-2630 Aartselaar
Tél.: +32-3-870 7300
Télécopie: +32-3-870 7301

Belorussia

Представительство ГРУНДФОС в
Минске
220123, Минск,
ул. В. Хоружей, 22, оф. 1105
Тел.: +(37517) 233 97 65,
Факс: +(37517) 233 97 69
E-mail: grundfos_minsk@mail.ru

Bosnia/Herzegovina

GRUNDFOS Sarajevo
Trg Heroja 16,
BiH-71000 Sarajevo
Phone: +387 33 713 290
Telefax: +387 33 659 079
e-mail: grundfos@bih.net.ba

Brazil

BOMBAS GRUNDFOS DO BRASIL
Av. Humberto de Alencar Castelo
Branco, 630
CEP 09850 - 300
São Bernardo do Campo - SP
Phone: +55-11 4393 5533
Telefax: +55-11 4343 5015

Bulgaria

Grundfos Bulgaria EOOD
Slatina District
Iztochna Tangenta street no. 100
BG - 1592 Sofia
Tel. +359 2 49 22 200
Fax. +359 2 49 22 201
email: bulgaria@grundfos.bg

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
50/F Maxdo Center No. 8 XingYi Rd.
Hongqiao development Zone
Shanghai 200336
PRC
Phone: +86-021-612 252 22
Telefax: +86-021-612 253 33

Croatia

GRUNDFOS CROATIA d.o.o.
Cebini 37, Buzin
HR-10010 Zagreb
Phone: +385 1 6595 400
Telefax: +385 1 6595 499
www.grundfos.hr

Czech Republic

GRUNDFOS s.r.o.
Čajkovského 21
779 00 Olomouc
Phone: +420-585-716 111
Telefax: +420-585-716 299

Denmark

GRUNDFOS DK A/S
Martin Bachs Vej 3
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Estonia

GRUNDFOS Pumps Eesti OÜ
Peterburi tee 92G
11415 Tallinn
Tel: + 372 606 1690
Fax: + 372 606 1691

Finland

OY GRUNDFOS Pumpat AB
Mestarintie 11
FIN-01730 Vantaa
Phone: +358-3066 5650
Telefax: +358-3066 56550

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS GMBH
Schlüterstr. 33
40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
e-mail: infoservice@grundfos.de
Service in Deutschland:
e-mail: kundendienst@grundfos.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706 / 27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Park u. 8
H-2045 Törökbálint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private Lim-
ited
118 Old Mahabalipuram Road
Thoraiappakkam
Chennai 600 096
Phone: +91-44 2496 6800

Indonesia

PT GRUNDFOS Pompa
Jl. Rawa Sumur III, Blok III / CC-1
Kawasan Industri, Pulogadung
Jakarta 13930
Phone: +62-21-460 6909
Telefax: +62-21-460 6910 / 460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit A, Merrywell Business Park
Ballymount Road Lower
Dublin 12
Phone: +353-1-4089 800
Telefax: +353-1-4089 830

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290 / 95838461

Japan

GRUNDFOS Pumps K.K.
Gotanda Metalion Bldg., 5F,
5-21-15, Higashi-gotanda
Shiagawa-ku, Tokyo
141-0022 Japan
Phone: +81 35 448 1391
Telefax: +81 35 448 9619

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul, Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Latvia

SIA GRUNDFOS Pumps Latvia
Deglava biznesa centrs
Augusta Deglava ielā 60, LV-1035, Rīga,
Tāl.: + 371 714 9640, 7 149 641
Fakss: + 371 914 9646

Lithuania

GRUNDFOS Pumps UAB
Smolensko g. 6
LT-03201 Vilnius
Tel: + 370 52 395 430
Fax: + 370 52 395 431

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

México

Bombas GRUNDFOS de México S.A. de
C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Netherlands
Veluwezoom 35
1326 AE Almere
Postbus 22015
1302 CA ALMERE
Tel.: +31-88-478 6336
Telefax: +31-88-478 6332
E-mail: info_gnl@grundfos.com

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Albany, Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznania
PL-62-081 Przeźmierowo
Tel: (+48-61) 650 13 00
Fax: (+48-61) 650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

România

GRUNDFOS Pompe România SRL
Bd. Biruintei, nr 103
Pantelimon county Ilfov
Phone: +40 21 200 4100
Telefax: +40 21 200 4101
E-mail: romania@grundfos.ro

Russia

ООО Грундфос
Россия, 109544 Москва, ул. Школьная
39
Тел. (+7) 495 737 30 00, 564 88 00
Факс (+7) 495 737 75 36, 564 88 11
E-mail
grundfos.moscow@grundfos.com

Serbia

GRUNDFOS Predstavništvo Beograd
Dr. Milutina Ivkovića 2a/29
YU-11000 Beograd
Phone: +381 11 26 47 877 / 11 26 47
496
Telefax: +381 11 26 48 340

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
24 Tuas West Road
Jurong Town
Singapore 638381
Phone: +65-6865 1222
Telefax: +65-6861 8402

Slovenia

SIA GRUNDFOS d.o.o.
Šlandrova 8b, SI-1231 Ljubljana-Črnuče
Phone: +386 1 568 0610
Telefax: +386 1 568 0619
E-mail: slovenia@grundfos.si

South Africa

Corner Mountjoy and George Allen
Roads
Wilbart Ext. 2
Bedfordview 2008
Phone: (+27) 11 579 4800
Fax: (+27) 11 455 6066
E-mail: lsmart@grundfos.com

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuentecilla, s/n
E-28110 Algete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
Box 333 (Lunnagårdsgatan 6)
431 24 Mölndal
Tel.: +46(0)771-32 23 00
Telefax: +46(0)31-331 94 60

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-1-806 8111
Telefax: +41-1-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
7 Floor, 219 Min-Chuan Road
Taichung, Taiwan, R.O.C.
Phone: +886-4-2305 0868
Telefax: +886-4-2305 0878

Thailand

GRUNDFOS (Thailand) Ltd.
92 Chaloom Phrakiat Rama 9 Road,
Dokmai, Pravej, Bangkok 10250
Phone: +66-2-725 8999
Telefax: +66-2-725 8998

Turkey

GRUNDFOS POMPA San. ve Tic. Ltd.
Sti.
Gebze Organize Sanayi Bölgesi
İhsan dede Caddesi,
2. yol 200. Sokak No. 204
41490 Gebze/ Kocaeli
Phone: +90 - 262-679 7979
Telefax: +90 - 262-679 7905
E-mail: satis@grundfos.com

Ukraine

ТОВ ГРУНДФОС УКРАЇНА
01010 Київ, Вул. Московська 86,
Тел.: (+38 044) 390 40 50
Факс.: (+38 044) 390 40 59
E-mail: ukraine@grundfos.com

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971-4- 8815 166
Telefax: +971-4-8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 8TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
17100 West 118th Terrace
Olathe, Kansas 66061
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

Uzbekistan

Представительство ГРУНДФОС в
Ташкенте
700000 Ташкент ул.Усмана Носира 1-й
тулик 5
Телефон: (3712) 55-68-15
Факс: (3712) 53-36-35

97622399 1110
Repl. 97622399 0410

ECM: 1067582

The name Grundfos, the Grundfos logo, and the payoff Be-Think-Innovate are registered trademarks owned by Grundfos Management A/S or Grundfos A/S, Denmark. All rights reserved worldwide.